STPM PHYSICS · SEMESTER 1 · CHAPTER 9

Deformation of Solids

9.1 Stress, Strain and Young's Modulus

DEFINITIONS & LAWS

- **Stress** is the force acting normally per unit cross-sectional area of a material.
- **Strain** is the ratio of the extension to the original length; it has no unit.
- **Young modulus** is the ratio of tensile (or compressive) stress to the corresponding strain, within the limit of proportionality.
- **Hooke's law**: the extension of a material is directly proportional to the applied force provided the limit of proportionality is not exceeded.
- **Limit of proportionality** is the point on the graph beyond which stress is no longer proportional to strain. **Elastic limit** is the maximum stress beyond which the material does not return to its original length when the force is removed.
- **Yield point** is the point at which the material extends considerably (plastic flow) with little or no increase in stress. **Ultimate tensile stress** is the maximum stress the material can withstand.
- A **ductile** material shows a large plastic region before fracture (copper, mild steel); a **brittle** material fractures just after the elastic region with almost no plastic deformation (glass, cast iron); a **polymeric** material (rubber) shows a large non-linear extension and hysteresis.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Stress	$\sigma = \frac{F}{A}$	σ = stress (Pa = N m^{-2}); F = normal force (N); A = cross-sectional area (m^2)
Strain	$\varepsilon = \frac{x}{L_0}$	ε = strain (no unit); x = extension (m); L_0 = original length (m)
Young modulus	$E = \frac{\sigma}{\varepsilon} = FL_0/(Ax)$	E = Young modulus (Pa); E is the gradient of the linear part of a σ–ε graph
Force constant	$F = kx;\ k = \frac{EA}{L_0}$	k = force constant (N m^{-1}); valid below the limit of proportionality; E = Young modulus (Pa); A = cross-sectional area (m^2); L_0 = original length (m)
Area of a circular wire	$A = \frac{\pi d^2}{4} = \pi r^2$	d = diameter (m); r = radius (m); A = cross-sectional area (m^2)
Extension of a wire under a given load	$x \propto \frac{L_0}{d^2}$	x = extension (m); L_0 = original length (m); d = diameter (m)

HOW TO ANSWER (MPM STYLE)

- Define stress and strain in words, then give the formula with symbols and units; write the strain as a pure number or percentage.
- Calculation: state $E = FL_0/(Ax)$ → convert mm to m, and use $A = \frac{\pi d^2}{4}$ with the diameter (not the radius) → substitute → answer with unit (Pa or N m^{-2}) to 3 s.f.
- Sketch of a σ–ε graph for a ductile metal: label axes (stress/Pa, strain), mark the limit of proportionality, elastic limit, yield point, ultimate tensile stress and fracture point; the straight line passes through the origin.
- Comparing materials: the steeper initial gradient means the larger Young modulus (stiffer); the larger maximum stress means stronger.

SECTION A TRAPS

Trap: Using the diameter as the radius in $A = \pi r^2$, or forgetting to convert mm^2 to m^2 ($1\ mm^2 = 10^{-6}\ m^2$); this gives an answer wrong by a factor of 4 or 10^6.

Trap: Confusing stiffness with strength: a stiffer material (larger gradient) is not necessarily the stronger one (larger ultimate tensile stress).

Trap: Reading E from a point on the curved part of the graph or from the area under the graph; E is the gradient of the initial straight line only.

TRICKS

Trick: For the same load and material, $x \propto \frac{L_0}{d^2}$. Doubling the diameter divides the extension by 4; doubling the length doubles the extension.

Trick: Unit check: Pa = N m^{-2} has the same dimensions as J m^{-3}, so energy density and stress share a unit.

9.2 Elastic Energy

DEFINITIONS & LAWS

- **Elastic potential energy (strain energy)** is the energy stored in a material when it is deformed by a force within the elastic limit.
- The work done in stretching a wire equals the area under the force–extension graph; this is stored as elastic potential energy if the elastic limit is not exceeded.
- **Energy density** is the elastic potential energy stored per unit volume of the material.
- Beyond the elastic limit, the work done is not wholly stored: part is dissipated as heat during plastic deformation, so the unloading line is not the loading curve (hysteresis for rubber: the area enclosed by the loop is the energy dissipated as heat).

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Elastic potential energy	$U = \frac{1}{2}Fx = \frac{1}{2}kx^2 = \frac{F^2}{2k}$	U = energy stored (J); F = force (N); x = extension (m); k = force constant (N m^{-1}); valid for the linear region only
Elastic energy in terms of Young modulus	$U = \frac{1}{2}\frac{EA}{L_0}x^2$	E = Young modulus (Pa); A = cross-sectional area (m^2); L_0 = original length (m); x = extension (m)
Energy density	$\frac{U}{V} = \frac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2E} = \frac{1}{2}E\varepsilon^2$	unit J m^{-3}; V = AL_0 = volume (m^3); σ = stress (Pa); ε = strain; E = Young modulus (Pa)
Work done for a non-linear curve	Work done = area under the F–x graph	count squares or use trapezium; F = force (N); x = extension (m)
Extra energy stored when load increases	$\Delta U = \frac{1}{2}(F_1 + F_2)(x_2 - x_1)$	for a linear graph; load increases from F_1 to F_2 (N); x_1, x_2 = corresponding extensions (m)

HOW TO ANSWER (MPM STYLE)

- Define elastic potential energy, then state $U = \frac{1}{2}Fx$ (area under the F–x graph) before substituting; convert extension to metres.
- For a stretched wire: find x from $x = \frac{FL_0}{AE}$ first, then $U = \frac{1}{2}Fx$; give the answer in J to 3 s.f.
- Energy conversion: equate the loss in gravitational potential energy (or kinetic energy) to $\frac{1}{2}kx^2$ and state the assumption that no energy is lost.
- For hysteresis or a non-linear graph, state that the energy equals the area under the curve or the area enclosed by the loop, as appropriate.

SECTION A TRAPS

Trap: Using U = Fx instead of $\frac{1}{2}Fx$; the factor ½ arises because the force increases from zero to F.

Trap: Using $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$ with the final values when the graph is curved (beyond the limit of proportionality); the area under the actual graph must be used.

Trap: Doubling the extension of the same wire multiplies U by 4 ($U \propto x^2$), not by 2; doubling the force also gives 4 times the energy.

TRICKS

Trick: For two springs or wires carrying the same force, $U = \frac{F^2}{2k}$: the smaller k stores more energy. With the same extension, $U = \frac{1}{2}kx^2$: the larger k stores more.

Trick: Unit check: $\frac{1}{2}\sigma\varepsilon$ has the unit Pa = J m^{-3}, so it is energy per unit volume, not total energy.